Practical 13 – Implement Your Own Ping Program

AIM

To create a basic ping utility that tests whether a host responds and measures the round-trip time.

ALGORITHM

1. Obtain a hostname or IP address from the user.
2. Convert the hostname to an IP address when necessary.
3. Open a raw socket that uses the ICMP protocol.
4. Prepare and transmit an ICMP Echo Request packet.
5. Wait for a response and calculate the elapsed time.
6. Print the response time or a timeout message, repeat the test, and close the socket.

PYTHON CODE

```
import socket
import struct
import time

host = input("Enter host name or IP address: ")
ip = socket.gethostbyname(host)
sock = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_ICMP)
sock.settimeout(2)

for i in range(4):
    packet_id = i
    packet = struct.pack("!BBHHH", 8, 0, 0, packet_id, 1)
    checksum = 0
    for j in range(0, len(packet), 2):
        checksum += (packet[j] << 8) + packet[j + 1]
    checksum = (checksum >> 16) + (checksum & 0xffff)
    checksum = ~checksum & 0xffff
    packet = struct.pack("!BBHHH", 8, 0, checksum, packet_id, 1)
    start = time.time()
    sock.sendto(packet, (ip, 0))
    try:
        sock.recvfrom(1024)
        end = time.time()
        print("Reply from", ip, "time =", round((end-start)*1000, 2), "ms")
    except socket.timeout:
        print("Request timed out")

sock.close()
```

INPUT

Enter a hostname or IP address, for example `8.8.8.8`.

OUTPUT

The program prints the reply time in milliseconds or displays “Request timed out” when no reply arrives.

NOTE

Raw ICMP sockets may need Administrator/root privileges. Some networks block ICMP, so a timeout alone does not prove that a host is offline.

RESULT

A simple ICMP ping program was created to send echo requests and report response times.